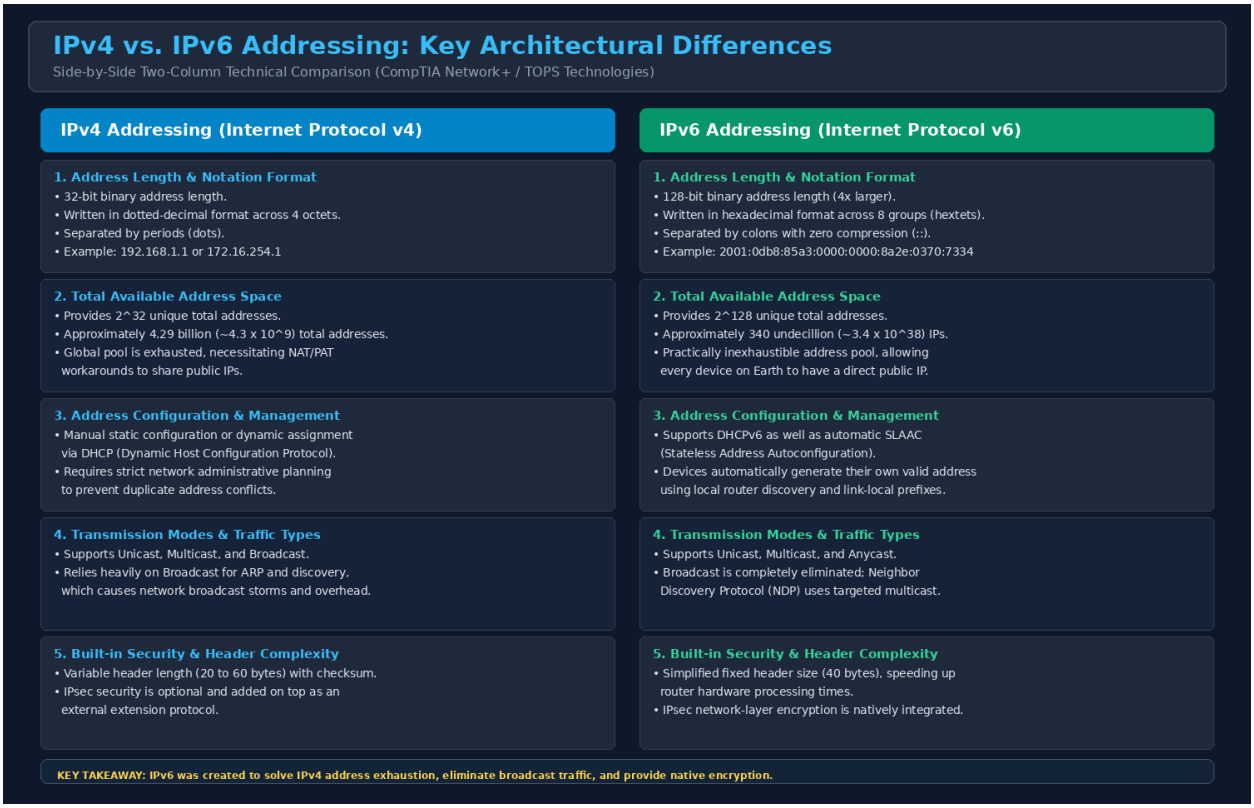
Introduction to IP Addressing

Task 1: Key Differences Between IPv4 and IPv6 Addressing



Task 2: Static IPv4 Address Assignment for Home Wi-Fi Network

- **Network:** 192.168.10.0/24
- **Subnet Mask:** 255.255.255.0
- **Default Gateway (Home Wi-Fi Router):** 192.168.10.1

Device Name	Assigned Static IP	Subnet Mask	Default Gateway	Purpose / Remarks
1. Laptop	192.168.10.10	255.255.255.0	192.168.10.1	Workstation client used for browsing and coursework.
2. Smartphone	192.168.10.20	255.255.255.0	192.168.10.1	Mobile station connecting via 5 GHz Wi-Fi.
3. Smart TV	192.168.10.30	255.255.255.0	192.168.10.1	Streaming display device for 4K video playback.
4. Network Printer	192.168.10.40	255.255.255.0	192.168.10.1	Static endpoint dedicated to wireless print jobs.

Task 3: Subnet Calculations for $192.168.1.0/27$

1. Number of Valid Subnets:

- Default Class C subnet mask = /24 (24 network bits).
- Given CIDR mask = /27 (27 network bits).
- Number of borrowed subnet bits (n):

$$n = 27 - 24 = 3\text{ bits}$$

- Total valid subnets created:

$$2^n = 2^3 = \mathbf{8\text{ valid subnets}}$$

2. Number of Usable Hosts per Subnet:

- Total bits in IPv4 = 32.
- Host bits remaining (h):

$$h = 32 - 27 = 5\text{ bits}$$

- Usable hosts per subnet:

$$2^h - 2 = 2^5 - 2 = 32 - 2 = \mathbf{30\text{ usable hosts per subnet}}$$

- *Block Size (Increment):* $2^5 = \mathbf{32}$.

3. First Three Valid Subnet Ranges:

Subnet Number	Network Address	Usable Host IP Range	Broadcast Address
Subnet 1	192.168.1.0	192.168.1.1 to 192.168.1.30	192.168.1.31
Subnet 2	192.168.1.32	192.168.1.33 to 192.168.1.62	192.168.1.63
Subnet 3	192.168.1.64	192.168.1.65 to 192.168.1.94	192.168.1.95

CIDR Subnetting Calculation: 192.168.1.0 with /27 Subnet Mask

Task 3: Step-by-Step Subnet & Host Mathematical Derivations + First Three Subnet Allocations

MATHEMATICAL FORMULAS & CALCULATION WORKING STEPS

Base Network & Default Class:	192.168.1.0 is a Class C network (Default mask: /24 or 255.255.255.0).
Given CIDR Subnet Mask:	/27 = 11111111 . 11111111 . 11111111 . 11100000 = 255.255.255.224
Borrowed Subnet Bits (n):	n = 27 - 24 = 3 bits borrowed from the 4th octet.
Host Bits Remaining (h):	h = 32 - 27 = 5 bits remaining for host addressing.
Total Number of Valid Subnets:	Formula: 2^n = 2^3 = 8 Valid Subnets.
Usable Hosts per Subnet:	Formula: 2^h - 2 = 2^5 - 2 = 32 - 2 = 30 Usable Hosts per Subnet.
Subnet Block Size (Increment):	Block Size = 256 - 224 = 32 (or 2^h = 2^5 = 32). Subnets increment by 32 in the 4th octet.

Subnet #	Network Address	Usable Host Range (First to Last)	Broadcast Address	Subnet Address Range
Subnet 1	192.168.1.0 /27	192.168.1.1 to 192.168.1.30	192.168.1.31	192.168.1.0 - 192.168.1.31
Subnet 2	192.168.1.32 /27	192.168.1.33 to 192.168.1.62	192.168.1.63	192.168.1.32 - 192.168.1.63
Subnet 3	192.168.1.64 /27	192.168.1.65 to 192.168.1.94	192.168.1.95	192.168.1.64 - 192.168.1.95

KEY TAKEAWAY: In each subnet of 32 total addresses, 2 addresses are reserved (Network ID & Broadcast ID), leaving exactly 30 usable host IPs.
Remaining subnets 4 to 8 increment sequentially: Subnet 4 (192.168.1.96), Subnet 5 (.128), Subnet 6 (.160), Subnet 7 (.192), Subnet 8 (.224).

Task 4: Subnet Analysis for $10.0.5.17$ with Mask $255.255.255.0$

Step-by-Step Calculation:

1. **Convert Subnet Mask to CIDR:**
    - $255.255.255.0$ has 24 contiguous binary ones:

$$11111111.11111111.11111111.00000000 = \mathbf{/24}$$

2. **Bitwise AND Operation for Network Address:**
    - Convert IP Address and Subnet Mask to binary:

Plaintext

IP: 10.0.5.17 -> 00001010.00000000.00000101.00010001

Mask: 255.255.255.0-> 11111111.11111111.11111111.00000000

AND: 10.0.5.0 -> 00001010.00000000.00000101.00000000

- **Network Address:** $\mathbf{10.0.5.0}$

3. **Determine Broadcast Address:**
    - Set all 8 host bits in the 4th octet to binary 1 ($11111111_2 = 255_{10}$):
    - **Broadcast Address:** $\mathbf{10.0.5.255}$

4. **Determine Usable Host Range:**
    - First Usable Host = Network Address $+ 1 = \mathbf{10.0.5.1}$
    - Last Usable Host = Broadcast Address $- 1 = \mathbf{10.0.5.254}$

Summary of Results:

- **Network Address:** 10.0.5.0
- **Broadcast Address:** 10.0.5.255
- **Range of Valid Host Addresses:** 10.0.5.1 to 10.0.5.254 (254 usable hosts)

Task 5: Online Subnet Calculator Verification & Screenshot

The verification calculated via subnet-calculator.com for IP 10.0.5.17 with subnet mask 255.255.255.0 is shown below:

Verification Statement:

My manual calculations completely matched the online subnet calculator's output, confirming a Network Address of **10.0.5.0**, a Broadcast Address of **10.0.5.255**, and an exact Usable Host Range of **10.0.5.1 to 10.0.5.254**.